
Uncovering Conceptual Blindspots in Generative Image Models Using Sparse Autoencoders

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Abstract

Despite their impressive performance, generative image models trained on large-scale datasets systematically fail to produce images with seemingly simple concepts—e.g., human hands or objects appearing in groups of four—that are certain to appear in the training data. These failure modes have so far been documented only anecdotally, which has left unresolved whether they are idiosyncratic anomalies or, as we show, structural limitations of these models. We resolve this question by introducing a systematic approach that identifies and characterizes "conceptual blindspots"—concepts present in the training data but absent or misrepresented in a model's generations. Our method leverages sparse autoencoders (SAEs) to extract interpretable concept embeddings, delivering a direct, quantitative comparison of concept prevalence between real and generated images. We train an archetypal SAE (RA-SAE) on DINOv2 features with 32,000 concepts—the largest such SAE to date—which enables fine-grained analysis of conceptual disparities at a resolution no prior method achieves. Applied to four popular generative models (Stable Diffusion 1.5/2.1, PixArt, and Kandinsky), our approach pinpoints specific suppressed blindspots (e.g., bird feeders, DVD discs, and whitespaces on documents) and exaggerated blindspots (e.g., wood background texture and palm trees). At the individual datapoint level, we further isolate memorization artifacts — instances where models demonstrably reproduce highly specific visual templates seen during training. Together, these results establish a theoretically grounded framework that systematically identifies conceptual blindspots in generative models by measuring their conceptual fidelity with respect to the underlying data-generating process.

1 Introduction

Generative image models trained on large scale datasets have achieved unprecedented capabilities, driving applications both within the vision domain [1-9] and well beyond it [9-15]. Yet this success masks a striking failure: a growing body of qualitative [16-18] and quantitative studies [19-20] establishes that these models cannot reliably generate even relatively simple concepts, e.g., human hands [21-24], objects appearing in groups of four [25], and negations or object relations [20, 26]. When prompted for such concepts, models reliably produce outputs with related structure but not the ground-truth concept itself—e.g., hands with six fingers. We term these failure modes "conceptual blindspots"¹ and they are genuinely counterintuitive: the models have unquestionably been exposed to abundant demonstrations that accurately detail these very concepts. This forces the central question of this work: are such failures isolated quirks of a few specific concepts, or the signature of a systematic phenomenon in which, across a broad spectrum of

¹We borrow the term "blindspots" from psychology literature [27], wherein it is used to refer to scenarios where an agent makes biased decisions despite exposure to observations that contradicts their decision's rationale.



Figure 1: $\delta(k)$ precisely quantifies a model’s tendency to over- or under-generate a concept c_k relative to its natural-data frequency. We classify concepts with $\delta(k) < 0.1$ as **suppressed conceptual blindspots** and those with $\delta(k) > 0.9$ as **exaggerated conceptual blindspots**. The depicted images, generated by four popular generative image models, showcase images exhibiting conceptual blindspots alongside aligned concepts.

concepts, models are consistently either unable or overly likely to produce images containing them? We show it is the latter.

This question is out of reach for existing approaches to evaluating generative image models [28, 29]. Existing approaches rely on coarse-grained measures of image realism, e.g., FID [30], and therefore cannot capture distributional failures at all. Methods like CLIPScore evaluate generation diversity or distribution-coverage statistics [31–33], offering at best partial insight, since they operate far above the granularity of fine-grained features or concepts [34, 35] and thus cannot localize conceptual blindspots. Qualitative analyses, such as participant surveys [5, 17, 36–38] or open-ended exploration [39], surface individual conceptual failures but provide no scalable methodology that can be repeated across model families and used to compare them. No existing tool answers the question we pose.

This work. We therefore develop the methodology this problem demands: an automated, unsupervised procedure that elicits precisely those concepts in the data distribution whose odds of generation by the true data-generating process diverge from those of the trained model. We make three contributions:

- **Formalizing Conceptual Blindspots in Generative Image Models.** We introduce the first systematic framework for identifying and quantifying conceptual blindspots in generative image models relative to natural images (Section 3). This formalization replaces anecdotal or human-defined concept evaluations with a principled account of the models’ limitations.
- **A Scalable, Unsupervised Approach for Identifying Conceptual Blindspots using Sparse Autoencoders.** We build a structured methodology that uses sparse autoencoders (SAEs) to extract and compare concept distributions between natural and synthesized images (Section 4). To this end, we train and open-source an archetypal SAE (RA-SAE) on DINOv2 features with 32,000 concepts—the largest such RA-SAE to date.
- **Exploratory Tool and Analysis.** Our exploratory web tool enables both distribution- and datapoint-level analysis of blindspots across models (Sections 5.1, 5.4). Applying our method to Kandinsky, PixArt, and Stable Diffusion (SD) 1.5/2.1 (Section 5), we pinpoint concrete instances of both suppressed conceptual blindspots (e.g., bird feeders, DVD discs, and whitespaces on documents) and exaggerated conceptual blindspots (e.g., wood background texture and palm trees), shown in Fig. 1.

The model, code, and web tool are available at a software repository.

2 Related Work

2.1 Explainability in Vision

Early work in explainable AI, including computer vision, focused on methods for attribution of influential input regions [40, 42]. These methods, however, convey little semantic information about learned representations and frequently produce outright incorrect explanations [43–45]. Concept-based interpretability [46] arose precisely to overcome this limitation, identifying semantically meaningful directions in neural networks and revealing not just where they look but what concepts and structures they employ [47–49].

Recent work establishes that the leading concept-based interpretability methods—ACE [44], CRAFT [48], and Sparse Autoencoders (SAEs) [50, 51]—all solve the same dictionary learning

task under different constraints [52]. Among these, sparse autoencoders (SAEs) are the most scalable route to concept-based interpretability. The original SAEs do carry known limitations—overly specific features [53], compositionality challenges [54], and limited intervention effects [55]—but improved variants directly address them, most notably the archetypal SAE (RA-SAE) [56] that we adopt.

Other interpretability methods—prompt-based probing [57], attention map or activation visualizations [58–59], and dataset-level statistics [32, 33] such as diversity or distribution-coverage metrics—yield only partial answers to these questions. They lack granularity by construction, operating on full images or prompts rather than fine-grained features and concepts [34, 35]. They further hinge on subjective interpretation and fail to distinguish between distinct failure modes [60]. Existing methods and metrics are therefore fundamentally inadequate for systematically identifying feature- and concept-level weaknesses of generative image models [28]—the gap our work closes.

2.2 Generative Image Models

Diffusion-based methods now dominate generative vision modeling across modalities, including image [4, 61–64], video [65–72], and 3D [7, 73–75]. For image generation, this line traces back to denoising diffusion probabilistic models (DDPMs) [76], later extended to non-Markov diffusion processes with denoising diffusion implicit models (DDIMs) [77].

The Stable Diffusion (SD) [9] model family made DDPMs highly accessible both in the research and open-source communities. The original SD was followed by several subsequent versions, including SD 2 [78], SD 3 [79], SD XL [80]. Many modifications and extensions of the SD architecture have emerged, enabling additional constraints for the diffusion process (e.g., style [81, 82], pose [83], and identity [84, 85]) as well as different input modalities, such as image-to-image generation. Different Latent Diffusion Models built on top of SD—including Kandinsky [86], PixArt [87], DeepFloyd [4, 88], and FLUX [89]—have also emerged.

2.3 Datasets for Generative Image Models

The recent success of generative vision models rests directly on the abundance of computational resources and large-scale internet datasets [90–92]. LAION-5B [93] in particular has been central to training open-source text-to-image models, including SD and its derivatives. Scraped from Common Crawl [94], it contains over 5B image-caption pairs, 2.3B of which are in English. Other prominent datasets include COYO-700M [95] and Conceptual Captions [96], with 700M and 3M image-caption pairs, respectively.

As LAION-5B gained popularity, its biases drew mounting scrutiny [97–101]. Filtering attempts notwithstanding, harmful content persisted [97–99], including NSFW material [100] and hundreds of CSAM instances [101], which forced its temporary removal from official channels. The dataset is further marred by low-quality images [102] and internet-style captions (e.g., product descriptions) that misalign with how users actually prompt trained models [103].

2.4 Concept Discovery and Sparse Coding in Generative Image Models

Dictionary learning seeks to find sparse representations of input data, where each sample can be reconstructed using a linear combination of few dictionary atoms [104–106]. Built upon compressed sensing theory [107, 108], the field evolved from early vector quantization methods [109] to sophisticated approaches including Non-negative Matrix Factorization [110, 111], Sparse PCA [112], and K-SVD [113]. Recent advances include online methods [114], structured sparsity [115], and theoretical guarantees [116, 117], alongside growing connections to deep learning [118, 119].

Advances in sparse coding have also been leveraged to study the emergence of high-level concepts inside diffusion models [120]. Prior to diffusion models, concept-grounded interpretability was applied to earlier generative architectures through concept-bottleneck models, which require human intervention at training time [121], and post-hoc detectors that retrofit concept supervision [122]. Both approaches, however, depend on human-defined concepts and therefore necessarily miss any broader trend the user does not explicitly anticipate.

Aggregate metrics (e.g., precision, recall, density, and coverage [35, 123]) and latent density scores, which predict sample quality from the model’s latent space [124], have also been proposed to evaluate

generative image models. These reveal distributional gaps but say nothing about which specific concepts are under- or over-represented.

These limitations make the case unavoidable for a scalable, unsupervised framework that systematically identifies and quantifies concept-level failure modes in generative image models [125]—exactly what we contribute.

3 Formalizing Conceptual Blindspots in Generative Models

We begin by formalizing the notion of *conceptual blindspots*: systematic discrepancies between the conceptual content of natural images and that of model-generated outputs. This formulation directly yields principled, quantitative measures that pin down exactly which concepts a generative model under- or over-represents relative to its data distribution. The process is illustrated in Fig. 2. We adopt standard assumptions in this pursuit [126, 130], and, importantly, find that our method elicits meaningful phenomenology even when these assumptions are violated.

Definition 1 (Data-Generating Process). Let $\mathcal{C} \subset \mathbb{R}^K$ denote a latent space with a Boltzmann prior $p(\mathbf{c}) = \exp(-E(\mathbf{c}))Z^{-1}$, where $E(\cdot)$ denotes an energy function that linearly decomposes over individual latents and Z is the corresponding partition function, i.e., $E(\mathbf{c}) = \sum_k E(c_k)$ and hence $p(\mathbf{c}) = \prod_k p_k(c_k)$. A data-generating process (DGP) is an invertible function $G : \mathcal{C} \rightarrow \mathcal{X}$ that maps the latents $\mathbf{c} \in \mathcal{C}$ to observations $\mathbf{x} \in \mathcal{X}$, i.e., $\mathbf{x} = G(\mathbf{c})$.

For notational simplicity, we use $p(\cdot)$ to denote both the latent density $p(\mathbf{c})$ and its push-forward to image space $p(\mathbf{x})$, where $\mathbf{x} = G(\mathbf{c})$. This is justified by the invertibility of G , which induces a valid distribution over $\mathcal{X}$ via the change of variables formula. In essence, the individual dimensions of the latent space reflect the **Concepts** underlying the data-distribution $P_{\mathcal{X}}$, defined over some observation space of images $\mathcal{X}$. For example, different latents may correspond to concepts like color, shape, size, location, and so on [131, 132]. We also let the data-generating process associate a text-description $\mathbf{t} \in \mathcal{T}$ with any image sampled from the data distribution, but do not explicitly model it. These text descriptions can then be used to train a text-conditioned **Generative image model** g_{θ} , with parameters θ , on a set of image-text pairs to map a noise signal $\eta \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and a text-description of the scene $\mathbf{t}$ to produce images $\mathbf{x}$ illustrating the latter.

To define conceptual blindspots in the model g_{θ} , we compare the probability mass a concept receives under the data-generating process against the mass the model assigns to it. To this end, we take an evaluation set of natural image-text pairs $(\mathcal{D}_{\mathcal{X}}, \mathcal{D}_{\mathcal{T}})$ and define a set of synthetically generated images $\mathcal{D}'_{\mathcal{X}}$ using the text descriptions. To estimate the probability of occurrence of a concept, we consider an **Energy model** $\xi : \mathcal{X} \rightarrow \mathbb{R}^d$ that maps images to a d -dimensional vector, where each dimension associates a scalar representing the energy in the k^{th} concept, i.e., $\xi_k(\mathbf{x}) = E(c_k)$. These estimates are feasible because we assume the data-generating process is invertible. Correspondingly, the probability the data-generating process associates with the image $\mathbf{x}$ can then be defined as $p_k(\mathbf{x}) \propto \exp(-\sum_k \xi_k(\mathbf{x}))$ (where Z_{ξ} is the partition function), hence yielding a population-level estimate $p_k(\mathcal{D}_{\mathcal{X}}) = \prod_{\mathbf{x} \in \mathcal{D}_{\mathcal{X}}} p_k(\mathbf{x})$. Using this and the sigmoid map $\sigma(\cdot)$, we define below the average energy difference in the k^{th} concept between the datasets $\mathcal{D}_{\mathcal{X}}, \mathcal{D}'_{\mathcal{X}}$.

Definition 2 (Energy Difference). Let $\mathbf{x} \in \mathcal{D}_{\mathcal{X}}$ denote a real image sampled from the data-generating process G , and let $\mathbf{x}' \in \mathcal{D}'_{\mathcal{X}}$ be a synthetic image generated by the model g_{θ} . Let $\xi_k : \mathcal{X} \rightarrow \mathbb{R}$ denote the energy assigned to the k^{th} concept by the energy model ξ . We define the

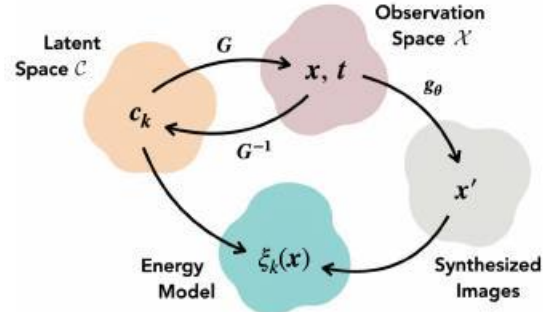


Figure 2: **Intuitive mapping of our framework.** Latent concepts $c_k \in \mathcal{C}$ map to observations $(\mathbf{x}, \mathbf{t})$ through the (invertible) DGP. S_{θ} generates images $\mathbf{x}'$ from $\mathbf{t}$. The energy model $\xi_k(\mathbf{x})$ extracts concept representations from both $\mathbf{x}$ and $\mathbf{x}'$, directly enabling the comparison of concept distributions that identifies blindspots.

179 energy difference for concept k as:

$$\begin{aligned}\delta_{g_\theta \leftrightarrow G}(k) &= \sigma(\mathbb{E}_{\mathbf{x}'}[\xi_k(\mathbf{x}')] - \mathbb{E}_{\mathbf{x}}[\xi_k(\mathbf{x})]) \\ &= \frac{p_k(\mathcal{D}'_{\mathcal{X}})}{p_k(\mathcal{D}_{\mathcal{X}}) + p_k(\mathcal{D}'_{\mathcal{X}})},\end{aligned}\tag{1}$$

180 where the expectations are taken over $\mathcal{D}'_{\mathcal{X}}$ and $\mathcal{D}_{\mathcal{X}}$, respectively, and $p_k(\mathcal{D}) \propto \exp(-\sum_{\mathbf{x} \in \mathcal{D}} \xi_k(\mathbf{x}))$
181 denotes the unnormalized conceptual probability mass of dataset $\mathcal{D}$ under concept k .

182 The energy difference in the k^{th} concept thus captures exactly the ratio of the probability that a
183 concept occurs in a set of observations (here, $\mathcal{D}'_{\mathcal{X}}$) to that in a baseline dataset (here, $\mathcal{D}_{\mathcal{X}}$). This
184 measure lets us define conceptual blindspots directly.

185 **Definition 3 (Suppressed / Exaggerated Conceptual Blindspots).** Given a generative image model
186 g_θ , we say, compared to the data-generating process G , c_k is a suppressed conceptual blindspot in
187 the model if $\delta_{g_\theta \leftrightarrow G}(k) < \lambda_{\min}$ and exaggerated if $\delta_{g_\theta \leftrightarrow G}(k) > \lambda_{\max}$.

188 In short, a conceptual blindspot is a concept whose likelihood of occurrence in generated images
189 departs sharply—through either suppression or exaggeration—from its prevalence under the data-
190 generating process. Suppressed concepts exhibit disproportionately low activation (e.g., $\delta(k) < \lambda_{\min}$),
191 whereas exaggerated concepts are overrepresented (e.g., $\delta(k) > \lambda_{\max}$). Throughout our analysis, we
192 set $\lambda_{\min} = 0.1$ and $\lambda_{\max} = 0.9$ to isolate these regimes.

193 This definition generalizes the classical notion of “mode collapse” (e.g., Bau et al. [39]); the decisive
194 difference is granularity. Mode collapse concerns exaggerated or suppressed odds of generating
195 entire images, whereas we resolve shifted odds at the level of specific concepts. For instance, when a
196 model fails to produce images of an object with a white background, we identify that concept as a
197 suppressed conceptual blindspot.

198 4 Method: Operationalizing the Definition of Conceptual Blindspots

199 We now present our pipeline for identifying conceptual blindspots in a generative model g_θ . Following
200 Sec. 3 this requires exactly two ingredients: (i) a set of images sampled from g_θ that support
201 comparison with the ground-truth generative process, and (ii) an energy model that enables that
202 comparison. Below, we use $\|\cdot\|_F$ to denote the Frobenius norm and $\|\cdot\|_0$ to denote the number
203 of non-zero entries (the ℓ_0 pseudo-norm). For a vector or matrix $\mathbf{X}$, $\mathbf{X} \geq 0$ implies element-wise
204 non-negativity. For $n > 0$, we let $[n] := \{1, \dots, n\}$, and denote the i -th row of a matrix $\mathbf{A}$ by $\mathbf{A}_i$.

205 **From Prompts to Latent Representations.** To identify conceptual blindspots in a model g_θ , we
206 compare a dataset $\mathcal{D}_{\mathcal{X}}$ of image-caption pairs $(\mathbf{x}, t)$ sampled from the data-generating process G and
207 their synthetic counterparts sampled from the generative model g_θ using the text descriptions.

208 Specifically, given t , we synthesize a counterpart image $\mathbf{x}'$ using a pretrained text-to-image generator
209 $g_\theta : \mathcal{T} \rightarrow \mathcal{X}$, implemented as a denoising diffusion probabilistic model (DDPM) [78, 86, 87].
210 Sampling occurs in latent space via a reverse trajectory $(\gamma_t)_{t=0}^T$:

$$\gamma_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad \gamma_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(\gamma_t - \frac{1-\alpha_t}{\sqrt{1-\alpha_t}} \epsilon_\theta(\gamma_t, t, c) \right) + \sigma_t \eta_t, \quad \text{and} \quad \eta_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I}),$$

211 where $\alpha_t \in (0, 1)$ and $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$ follow the standard cosine noise schedule. The final latent γ_0
212 is decoded via a pretrained VAE to yield the synthetic image $\mathbf{x}' = \text{VAE}(\gamma_0)$. For the remainder of
213 the paper, we treat g_θ as a black box that maps prompts to images: $t \mapsto \mathbf{x}'$.

214 **Defining the Energy Model.** Prior work establishes that self-supervised learning methods invert the
215 data-generating process and recover the underlying energy function up to linear transformations [126
216 128, 133, 134]; we build directly on this result and adopt DINOv2 [135], a state-of-the-art self-
217 supervised model, for our analysis. Since the number of concepts underlying the DGP exceeds
218 the dimensionality of the model’s feature space [51, 136], we train sparse autoencoders (SAEs) on
219 its features to isolate the subspaces corresponding to these concepts [50, 56, 137–139]. The logic
220 is direct: if DINOv2 encodes the concepts underlying the generative process along approximately
221 orthogonal directions (as our independence constraint in Def. 1 assumes), then an SAE isolates these
222 concepts along individual dimensions of its latent space [136]. The activation the SAE assigns to

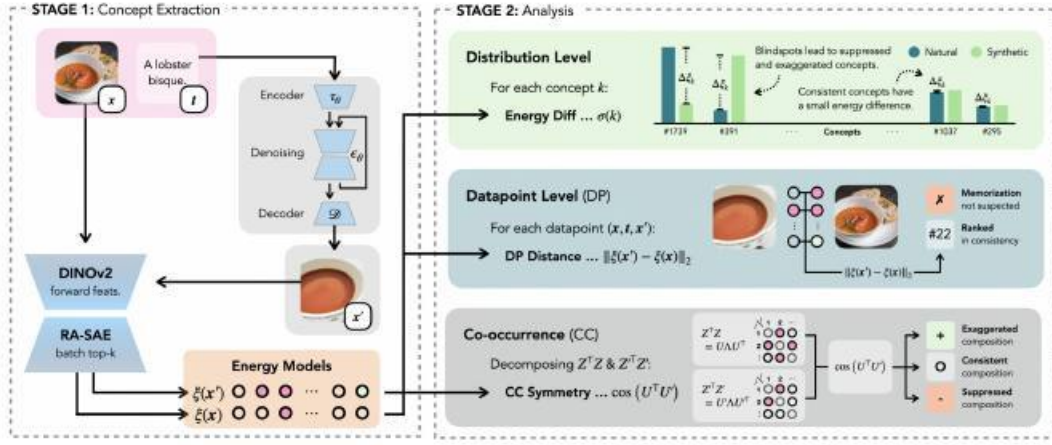


Figure 3: **Concept Extraction Pipeline.** For a triplet (x, t, x') , we extract the concepts in x and t from each image’s DINOv2 features, which a RA-SAE maps into sparse concept embeddings, yielding the energy models $\xi(x)$ and $\xi(x')$. In particular, $\xi_k(x) = E(c_k)$ holds the energy in the k^{th} concept.

each dimension then serves as our approximation of the ground-truth energy function of the concept that dimension models.

Formally, using $f : \mathcal{X} \rightarrow \mathbb{R}^d$ to denote our feature extraction module (i.e., the DINOv2 model), we extract features $a = f(x) \in \mathbb{R}^d$ from both natural and synthetic images from datasets $\mathcal{D}_\mathcal{X}, \mathcal{D}'_\mathcal{X}$. Assuming the count of image-text pairs is n , we stack the real and generated features into matrices $A, A' \in \mathbb{R}^{n \times d}$. We then decompose each feature vector into a sparse combination of learned concept atoms using an SAE. Specifically, let $D \in \mathbb{R}^{d \times K'}$ denote a dictionary of K' concept vectors, and let $\Psi : \mathbb{R}^d \rightarrow \mathbb{R}^{K'}$ be the SAE encoder that maps input features to sparse codes. Applying Ψ row-wise yields the matrix of activations $Z = \Psi(A) \in \mathbb{R}^{n \times K'}$, where each row $z_i = \Psi(a_i)$ represents the concept decomposition of an image. The decoder reconstructs features via $ZD^\top$, and the SAE is trained to minimize the reconstruction error subject to sparsity and non-negativity:

$$\min_{\Psi, D} \|A - \Psi(A)D^\top\|_F^2 \quad \text{s.t.} \quad \Psi(A) \geq 0, \|\Psi(A)_i\|_0 \ll K' \quad \forall i \in [n]. \quad (2)$$

Vanilla SAEs drift toward arbitrarily oriented dictionaries, rendering downstream analyses acutely sensitive to the random seed. To eliminate this instability and make our study fully reproducible and seed-independent, we employ the *Archetypal* SAE (RA-SAE) [56] with a TOP-K sparsity constraint [137]. RA-SAE constrains the dictionary D to be a convex combination of training data. Specifically, we write $D = WA$ with $W \in \Omega_{K', n}$, the set of row-stochastic matrices in $\mathbb{R}^{K' \times n}$:

$$\Omega_{K', n} := \{W \in \mathbb{R}^{K' \times n} \mid W \geq 0, W\mathbf{1} = \mathbf{1}\}. \quad (3)$$

Every atom D_i therefore lies in the convex hull of the data $\text{conv}(A)$, and any reconstruction $ZD^\top$ resides inside the conic hull of the data $\text{cone}(A)$. This guarantees that learned concepts remain faithful to the support of the data distribution [56]. Once trained, the SAE delivers a consistent set of sparse codes: Z for real images and Z' for their generated counterparts. *These codes capture the same prompt-conditioned visual semantics in terms of shared, interpretable concepts, and the activation value of each concept furnishes the energy values that drive our analysis of conceptual blindspots.* Our method thus constitutes a structured pipeline that, given a prompt and its associated real image (t, x) , produces two sparse concept vectors (z, z') and enables direct comparison of real and generated visual content in a common conceptual basis.

To summarize the full pipeline: starting from a captioned image (t, x) , we synthesize a counterpart x' and map both images into a shared, sparse concept space via a vision encoder and a trained SAE, yielding interpretable representations (z, z') that form the foundation for evaluating the conceptual shifts the generative process induces.

5 Results

We analyze four generative image models trained on LAION-5B—SD 1.5, SD 2.1, PixArt, and Kandinsky—using $|\mathcal{X}| = 10,000$ image-text pairs and their corresponding generations (Fig. 4). Our analysis operates at three levels (Fig. 3): a **distribution-level** evaluation exposes suppressed and exaggerated concepts; a **datapoint-level** analysis pinpoints failures tied to ambiguity, omission, and memorization; and a **compositional** analysis uncovers subtle but systematic distortions in concept co-occurrence geometry.

Our core contribution is an interactive exploratory tool, shown in Fig. 5. The tool is a web-based interface built around a UMAP projection of concept representations that visualizes and compares concept-level energy differences. It is publicly available at a software repository, together with pre-computed energy difference data for the four models evaluated in this work (SD 1.5, SD 2.1, PixArt, and Kandinsky). Every subsequent analysis in this paper stems directly from insights this tool surfaces. Its primary functionalities, which power these analyses, include:

- **Contrast different models and architectures.** For each evaluated model, the tool provides a UMAP visualization spanning all 32,000 RA-SAE concepts. Each scatter point is an individual concept, color-coded by its energy difference.
- **Inspect concepts.** Every concept carries a card with key statistics, representative real and generated images (x , x'), and visualized co-occurrence patterns.
- **Explore blindspots.** Beyond the UMAP and per-concept views, the tool provides global rankings of suppressed and exaggerated blindspots, immediately surfacing the most notable conceptual blindspots.

Rather than exhaustively studying a single phenomenon, we present high-level findings that demonstrate the tool’s versatility and open the door to broader, customizable exploration.



Figure 4: **Representative image-caption pairs** from LAION-5B, paired with images generated from the same prompt by SD 1.5, SD 2.1, PixArt, and Kandinsky.

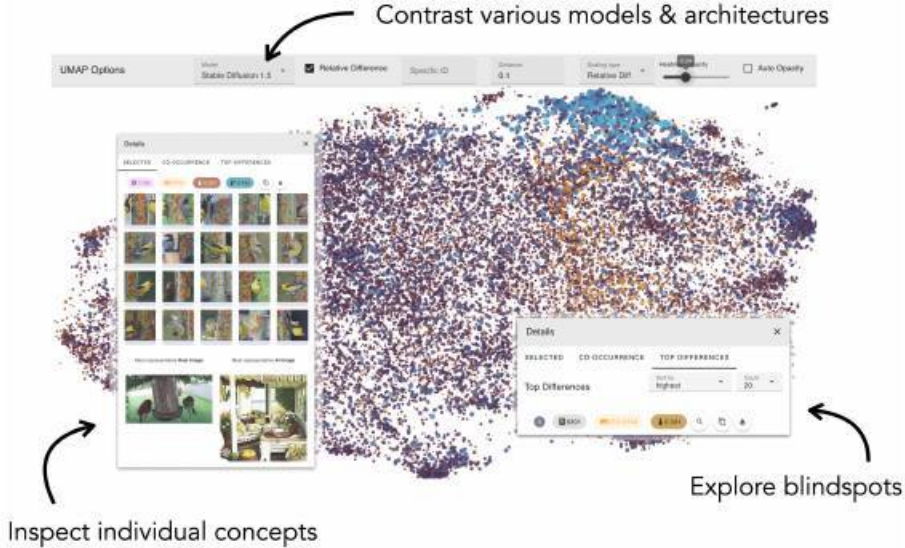


Figure 5: **Overview of the Exploratory Tool.** The web interface displays a UMAP projection for each evaluated model, where each dot is a concept color-coded by its energy difference. Selecting a concept opens a detail panel with illustrative images, statistics, and the most representative natural and generated images x and x' . An ordered list of the concept’s co-occurrences appears alongside global rankings of blindspots.

281 5.1 Generative Image Models Cannot Produce Unseen Concepts

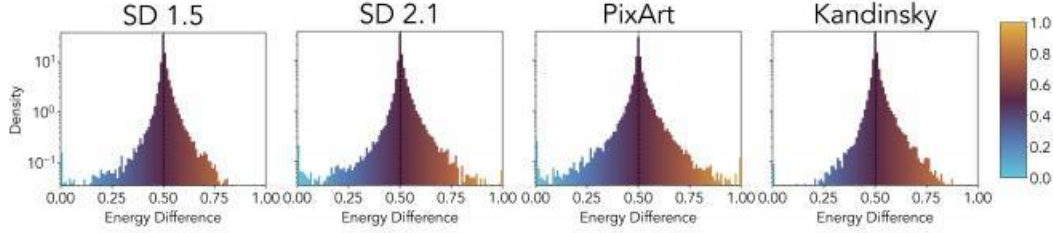


Figure 6: **Concept Energy Distribution.** Log-scale histograms of energy differences $\delta(k)$ across 32,000 concepts, comparing the natural and synthesized distributions for each evaluated model. Values left of zero mark suppressed concepts (under-represented); values right of zero mark exaggerated concepts (over-represented). The pronounced left-heavy asymmetry is consistent across all four models.

282 To quantify the disparities between the generative models and the natural image distribution, we first
 283 analyze the marginal energy difference $\delta(k)$ across the 32,000 concepts learned by RA-SAE. As
 284 defined in Sec. 4, this quantity measures the relative prevalence of each concept in the synthesized
 285 versus natural image sets: $\delta(k) < 0.1$ marks concept k as under-represented (suppressed) in the
 286 generated images, while $\delta(k) > 0.9$ marks it as over-represented (exaggerated). Fig. 6 presents
 287 the distribution of $\delta(k)$ for each of the four evaluated models. Every model exhibits heavy-tailed
 288 histograms with substantial mass at both extremes, establishing systematic discrepancies in concept
 289 coverage. Crucially, the left tail—corresponding to suppressed concepts—is consistently denser and
 290 longer than the right, evidence of a pervasive tendency toward concept suppression. This asymmetry
 291 is confirmed by the negative skewness of every distribution: Skewness = -0.54 for SD 2.1, -0.40
 292 for both SD 1.5 and PixArt, and -0.23 for Kandinsky.

293 While all models exhibit both suppressed and exaggerated concepts, their specific profiles differ.
 294 PixArt, for instance, shows a markedly wider spread, indicating a more suppressed concept distri-
 295 bution. The consistent left-skew across all distributions nonetheless establishes a shared tendency
 296 toward concept omission; we dissect the specific characteristics of this behavior in the following
 297 Sections.

298 5.2 Structure and Specificity of Conceptual Blindspots

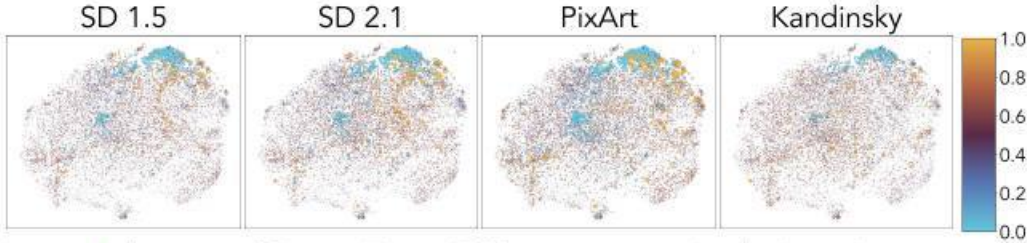


Figure 7: **Structure of Concept Energy Differences.** UMAP visualizations of 32,000 concepts, colored by their energy difference $\delta(k)$ between the natural and synthesized distributions. Clusters expose clear, structured patterns of conceptual blindspots, with suppressed concepts on the blue end and exaggerated ones on the yellow end.

299 Having quantified the marginal discrepancies in concept frequency, we now expose their global
 300 structure by embedding the full set of 32,000 concepts into two dimensions using UMAP on the
 301 sparse codes, coloring each concept by its $\delta(\cdot)$ value. Fig. 7 reveals distinct concept clusters across
 302 all models. These clusters coincide with contiguous regions of conceptual blindspots, especially for
 303 suppressed (blue) concepts, demonstrating that blindspots are highly structured rather than random—a
 304 signature of shared biases in either training distributions or architectural priors. To establish how
 305 consistent these blindspot patterns are across models, we analyze both the magnitude and structure of
 306 concept-level $\delta(\cdot)$ values. Fig. 8 presents scatter plots and pairwise Pearson correlation coefficients
 307 between the $\delta(k)$ vectors of SD 1.5 and every other model. SD 1.5 and 2.1 correlate strongly
 308 ($r = 0.82$), exactly as their shared architectural and training pipelines predict. Their correlations with

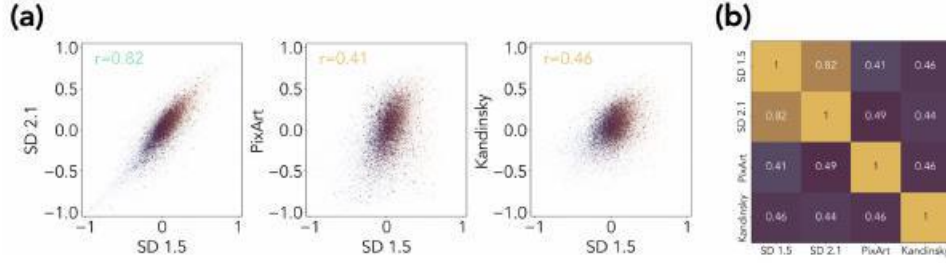


Figure 8: **Cross-Model Concept Energy Correlation.** Pairwise scatter plots of $\delta(k)$ across all four evaluated models, with Pearson correlation coefficients reported top left. The strong alignment between SD 1.5 and SD 2.1 stands in sharp contrast to the far weaker correlations among other architectures, confirming the presence of model-specific blindspots. **Correlation Matrix of Conceptual Blindspots.** Heatmap of pairwise Pearson correlation coefficients for δ between all models, quantifying the degree of shared conceptual blindspots across them.

PixArt and Kandinsky, by contrast, are far lower— $r = 0.41$ and $r = 0.46$, respectively—showing that these models emphasize genuinely different regions of the conceptual space.

This analysis establishes that blindspots fall into two distinct classes: some are universally shared—driven by properties of the dataset—while others are strongly model-specific, arising from idiosyncrasies in training dynamics or model capacity. This dichotomy motivates the separate study of shared and model-specific blindspots that we pursue in the subsequent sections.

5.3 Qualitative Examples of Blindspots

We next visualize specific examples of both suppressed and exaggerated blindspots to make concrete which concepts fall into these regimes. In Fig. 9a we show a conceptual blindspot suppressed by all models: every evaluated model fails to reproduce the concept **solid white on documents**. As the figures make plain, despite the caption explicitly referencing this concept, none of the generated images reflect the intended visual semantics—direct evidence that this region of the concept space is systematically under-sampled across models. Fig. 9b then isolates a model-specific blindspot: three models capture the concept **pan** accurately, yet it is conspicuously absent from Kandinsky’s generations. This directly corroborates Sec. 5.2 where cross-model agreement proved high for some concepts and limited for others.

Conversely, Fig. 10 presents a case of exaggeration, where the concept **shadow under animal** is markedly over-emphasized in generated images. Shadows are themselves plausible, but their consistent, pronounced rendering across models—set against the far more nuanced and variable occurrences in natural images—reveals an overactive prior. Notably, despite a deliberate search for concepts uniquely exaggerated by a single model, we found none, indicating that exaggerations are essentially universal across architectures.

Taken together, these examples concretely demonstrate how conceptual blindspots man-



Figure 9: **Examples of Suppressed Conceptual Blindspots.** The universal blindspot appears in all evaluated models; the unique blindspot appears only in Kandinsky.

Figure 10: **Example of an Exaggerated Conceptual Blindspot.** The universal blindspot appears in all evaluated models; the unique blindspot appears only in Kandinsky.

ifest in generated outputs and establish that our energy-based diagnostic reliably surfaces both shared and model-specific failure modes. It exposes surprising model limitations—such as the consistent, cross-model failure to reproduce clear or solid background elements like whitespaces on documents. Such patterns point to failure modes rooted in architectural constraints or training-data biases that transcend individual model idiosyncrasies.

These aggregate-level analyses are informative, but they raise a sharper question: do blindspots emerge only in aggregate across many samples, or do they persist at the level of individual datapoints? Answering it lets us probe the mechanisms behind blindspots directly—uncovering cases of prompt misinterpretation, latent memorization, or both.

5.4 Investigating Datapoint-level Energy Difference: From Incongruent to Memorized Images

To move beyond population-level statistics, we examine the individual natural vs. generated image pairs whose $\delta(\cdot)$ values, averaged across all concepts, are largest and smallest. This analysis cleanly separates model successes from failures, and it shows that the failures arise predominantly from prompt ambiguity or memorization artifacts. Fig. 11a shows instances with near-zero difference in average $\delta(\cdot)$ values. Here the generated images are conceptually indistinguishable from the original. Yet qualitative inspection makes clear that this happens not because the model faithfully captures the prompt semantics, but through pure replication of memorized templates: the repetitive visual structures (e.g., outlines of clothing or object arrangements) betray copying of overly frequent patterns from the training data. By contrast, Fig. 11b shows samples among the largest $\delta(\cdot)$ values. These mark pronounced conceptual divergence between the synthesized and natural image. Some discrepancies trace to underspecified or noisy captions, but others expose genuine blindspots: the prompt describes a clear concept faithfully present in $\mathcal{D}_X$, yet the model fails to realize it in $\mathcal{D}'_X$. This failure demonstrates that even with adequate language grounding, certain concepts lie squarely outside the model’s generative reach.



Figure 11: • **Datapoint-level Conceptual Alignment.** (a) Examples with minimal energy differences, where models memorize training patterns. (b) Examples with large differences, where pronounced concept divergences arise from prompt ambiguity or model limitations.

5.5 Analyzing Post-Training Effects

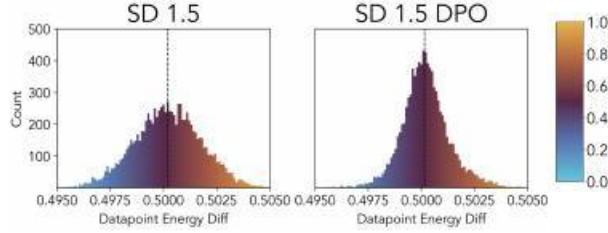


Figure 12: **Effect of DPO on Concept Fidelity.** Histogram of datapoint-wise energy differences between the synthesized and natural distributions of SD 1.5 with and without DPO. DPO shifts the distribution toward lower and tighter energy differences, evidencing improved conceptual alignment.

Post-training protocols such as safety fine-tuning have been argued to reduce the diversity of model generations [140]. Because our pipeline isolates precisely the differences between a model’s generations and the ground-truth DGP, we deploy it to characterize the effects of DPO, a popular safety fine-tuning protocol [141]. We compare two variants of the SD 1.5 model: one trained with DPO and one without. For each image pair $(\mathcal{D}_X, \mathcal{D}'_X)$, we compute the ℓ_2 norm of the difference between their internal concept energy vectors, $\|\xi(\mathcal{D}'_X) - \xi(\mathcal{D}_X)\|_2$. Fig. 12 presents a histogram of these datapoint-wise energy differences. The DPO-enhanced model exhibits both a lower median and a narrower spread, demonstrating that its generated concepts align more consistently with the ground-truth DGP. DPO therefore regularizes the model’s concept distribution, driving outputs to more closely reflect the semantic content of the seen inputs. While our analysis does not disentangle the specific inductive biases DPO introduces, these results provide direct empirical evidence that its optimization objective—favoring human-preferred generations—indirectly improves the match to the training distribution. In particular, it reduces *both over- and under-activation* of individual concepts relative to the natural baseline. These findings establish the value of our pipeline for characterizing the downstream effects of post-training interventions: not merely in terms of output quality, but in how they reshape the conceptual geometry of the model’s output space.

5.6 Conceptual Misalignment as a Function of Empirical Frequency

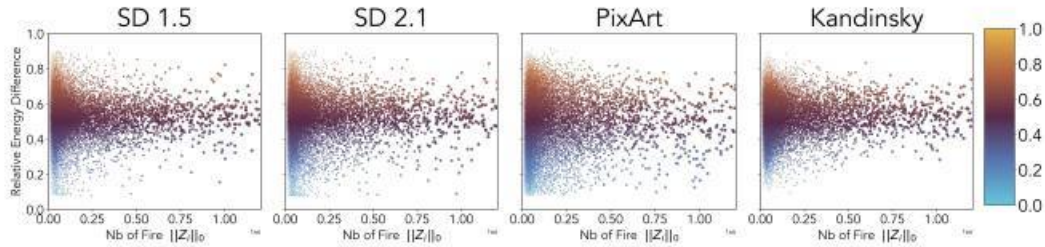


Figure 13: **Concept Fidelity Across the Frequency Spectrum.** Scatter plots of the relationship between concept frequency (x-axis) and energy difference (y-axis) across the four evaluated models. Each point is a concept, with size proportional to its activation frequency. Rare concepts consistently incur the largest energy differences.

We hypothesized above that conceptual blindspots are not merely architectural artifacts but also emerge as a direct consequence of the distributional peculiarities of certain features. We now test this hypothesis directly by examining whether concepts rarely activated in natural images—those in the long tail of the data distribution—give rise to blindspots in generative models. We process the natural dataset $\mathcal{D}_X$ through the trained SAE and compute, for each concept k , its empirical frequency $\|Z_{:,i}\|_0$, where $Z_{:,i}$ holds the activations of concept i across all our images. We then correlate this frequency with the absolute energy difference observed across generated outputs. Fig. 13 visualizes this relationship for all evaluated models. The pattern is clear and consistent: high-frequency concepts show low energy discrepancies, whereas rare concepts—especially suppressed ones ($\delta(k) < 0.5$)—exhibit large alignment errors. This establishes that many blindspots stem not from randomness or model quirks but from systematic effects tied to long-tail concept distributions.

411 Addressing them will require not only architectural changes but also targeted strategies such as data
412 reweighting or augmentation.

413 6 Discussion

414 Our analysis uncovers numerous conceptual blindspots across four popular generative image mod-
415 els. These results are only the beginning: each individual finding could anchor its own dedicated
416 investigation. Rather than exhaust any one of them, we demonstrate the breadth of our method and
417 exploratory tool. Out of the box, they systematically identify concepts that models fail to generate,
418 detect memorization artifacts, surface datapoints with insufficient captions, quantify post-training ef-
419 fects, and characterize conceptual shifts across model architectures. This establishes a foundation for
420 follow-up work to extend the depth of analysis, broaden the set of evaluated architectures, and pursue
421 the root causes of conceptual blindspots. A natural next step is to exploit hierarchical representations
422 of concepts in RA-SAE for still finer-grained analysis.

423 **Limitations.** We note two clear boundaries of the current instantiation. First, because we rely on
424 DINOv2 and RA-SAE for concept extraction and representation, our analysis covers exactly the
425 concepts these models capture; concepts they represent poorly fall outside its scope. Second, while
426 our sample of 10,000 images is substantial, it does not fully cover the long tail of rare concepts,
427 concept co-occurrence, or other compositional statistics. Both are constraints of the present setup
428 rather than of the framework itself, and each points to a concrete extension.

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Appendix

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Table of Contents

836

A Experimental Setup	23
-----------------------------	-----------

837

A.1 Observation Space	23
-----------------------	----

838

A.2 Synthesized Images	23
------------------------	----

839

A.3  Distribution Level Analysis	23
---	----

840

A.4  Datapoint Level Analysis	24
--	----

841

A.5  Co-occurrence Analysis	24
--	----

842

B Custom RA-SAE Training Details	25
---	-----------

843

C Computational Resources	26
----------------------------------	-----------

844

D Autointerpretability	27
-------------------------------	-----------

845

E Additional Results: Model-Specific Blindspots	29
--	-----------

846

F Additional Results: Higher-order Blindspots with Compositional Discrepancy	31
---	-----------

847

G Additional Results: Blindspot Stress Testing	33
---	-----------

848

G.1 Bird Feeder Blindspot in Kandinsky	33
--	----

849

G.2 Glossy DVD Disc Blindspot in SD 1.5	34
---	----

850

H Additional Examples: Synthesized Images	35
--	-----------

851

I Concentration bounds for δ	36
---	-----------

852

J Monotonicity and Calibration-Free Interpretation of δ	37
--	-----------

853

K Stability of FID Under SAE Embeddings	38
--	-----------

854

855

856

A Experimental Setup

This section details the experimental setup underlying our analysis of four popular generative image models: SD 1.5/2.1, Kandinsky, and PixArt, all trained on LAION-5B or its subsets/derivatives. The code is available at a software repository.

A.1 Observation Space

We construct the observation space by sampling 10,000 image-text pairs from the LAION-5B dataset [93], which serves as our domain of natural images. Because of CSAM and other unsafe content, the original release is no longer available; we therefore use a substitute release of a subset of this dataset with additional filtering of unsafe content, available at <https://huggingface.co/datasets/laion/relaion2B-en-research-safe>.

The sampling procedure is threefold: (1) load the full LAION dataset using the Hugging Face datasets library, (2) validate proper URL structure and resource availability via HTTP HEAD requests, and (3) sample randomly with replacement until reaching the target of 10,000 valid samples. This yields D_G as (x, t) tuples and guarantees an observation space of accessible image-text pairs, even though a large portion of the original image URLs has been made unavailable since release. Additional examples of synthesized images appear in Appendix H.

A.2 Synthesized Images

For each of the four evaluated models, we generate a synthetic dataset D_{g_θ} in one-to-one correspondence with D_G , yielding triplets (x, x', t) . Given the 10,000 image-text pairs (x, t) from D_G , we use each caption t to synthesize a counterpart image x' with every generative model g_θ .

The synthesis follows the standard text-to-image generation pipeline for each architecture, implemented with the Hugging Face diffusers library and models loaded at mixed precision (fp16). All synthetic images are generated at 512×512 pixel resolution with default parameters.

Stable Diffusion 1.5. The checkpoint from <https://huggingface.co/benjamin-paine/stable-diffusion-v1-5> (which is a mirror of the deprecated <https://huggingface.co/runwayml/stable-diffusion-v1-5>) is used. Inference is performed using 50 inference steps, with the guidance scale fixed at 7.5.

Stable Diffusion 1.5 + DPO. The DPO variant of SD 1.5 (used in the analysis in Sec. 5.5) follows the baseline SD 1.5 implementation, but replaces the UNet component with a DPO-trained version from <https://huggingface.co/mhdang/dpo-sd1.5-text2image-v1>.

Stable Diffusion 2.1. The checkpoint from <https://huggingface.co/stabilityai/stable-diffusion-2-1> is used. Inference is performed using 50 inference steps, with the guidance scale fixed at 7.5.

Kandinsky. The checkpoint from <https://huggingface.co/kandinsky-community/kandinsky-2-1> is used. Inference is performed using 100 inference steps, with the guidance scale fixed at 4.0.

PixArt. The checkpoint from <https://huggingface.co/PixArt-alpha/PixArt-XL-2-1024-MS> is used. Inference is performed using 50 inference steps, with the guidance scale fixed at 7.5.

A.3 Distribution Level Analysis

Section 5.1. We compute energy differences $\delta(\cdot)$ across all 32,000 concepts for each evaluated model. The sigmoid transformation with temperature $T = 0.8$ is applied during normalization. The resulting values are visualized as log-scale density histograms with 100 bins spanning $[0, 1]$.

Section 5.2. We embed the complete set of 32,000 concepts into two-dimensional space using UMAP applied to the sparse concept codes. Each point is an individual concept, colored by its energy difference $\delta(\cdot)$ to emphasize both suppressed and exaggerated blindspots. To quantify cross-model

consistency, we compute pairwise Pearson correlation coefficients between the $\delta(\cdot)$ vectors of all model pairs, producing both scatter plots and correlation matrices. This analysis establishes whether blindspots cluster in conceptual space and cleanly separates model-specific from universal patterns of conceptual blindspots.

Section 5.3 We rank all 32,000 concepts by their energy difference $\delta(\cdot)$ and manually examine the extrema (both suppressed and exaggerated blindspots), selecting concepts with $\delta(\cdot) < 0.1$ for suppressed blindspots and $\delta(\cdot) > 0.9$ for exaggerated ones. We annotate the presented examples with textual descriptions of each concept by inspecting its most activating images and spatial attention patterns. We describe our ongoing efforts to automate this concept interpretation in Appendix D

Section 5.5 We compare SD 1.5 with and without DPO as follows: for each image pair (x, x') , we compute the L2 norm of the difference between their concept energy vectors $\|\xi(x') - \xi(x)\|_2$, applying a sigmoid transformation with temperature $T = 0.8$ to the element-wise differences before taking their mean. This yields datapoint-wise energy differences that quantify precisely how much each generated image deviates from its natural counterpart in concept space. We then visualize these differences as overlapping histograms contrasting the two model variants.

Section 5.6 For each concept c_k , we count its empirical frequency $\|Z_{:,k}\|_0$ (the number of non-zero activations across the natural dataset) and apply a sigmoid normalization with temperature $T = 0.4$ to the energy differences $\delta(\cdot)$. We visualize the analysis as scatter plots with empirical concept frequency on the x-axis and the sigmoid-transformed energy difference on the y-axis; point sizes are proportional to activation frequency and point colors to the magnitude of energy differences.

A.4 • Datapoint Level Analysis

Section 5.4 For each image pair (x, x') , we compute the L2 norm of the difference between their concept energy vectors $\|\xi(x') - \xi(x)\|_2$, giving a scalar measure of conceptual divergence per pair. We then rank the samples by this energy difference. Minimal divergence flags memorization artifacts, while maximal divergence pinpoints significant conceptual failures. This ranking drives the qualitative inspection of specific failure modes.

A.5 • Co-occurrence Analysis

Appendix F For both the natural and synthesized data D_G and D_{g_θ} , we analyze concept co-occurrence patterns through the co-activation matrix $Z^T Z$, which holds the pairwise correlations in concept usage. We perform spectral analysis via eigendecomposition to examine the dominant conceptual directions, and assess the alignment between natural and synthetic co-occurrence structures using cosine similarity heatmaps between the top-100 eigenvectors of each co-occurrence matrix. We visualize these 100×100 similarity matrices as square heatmaps, in which perfect diagonal alignment indicates identical principal concept axes and off-diagonal patterns expose rotations and mismatches in compositional geometry.

938 B Custom RA-SAE Training Details

939 This section details the training of our custom archetypal SAE (RA-SAE) [56]. The model is
940 open-sourced at a software repository.

941 **Dataset.** The auto-encoder is trained on the complete ImageNet-1k training split, (≈ 1.28 M)
942 RGB images. Each image is converted to 261 visual tokens using DINOv2 [135]; tokens are fed
943 to the SAE without class or position embeddings. The total number of training tokens is therefore
944 $50 \times 1.28\text{M} \times 261 \approx 1.67 \times 10^{10}$.

945 **Dictionary.** The dictionary has 32,000 concept dimensions. For the sparse activation rule, top- k
946 masking with $k = 5$, is used; activations outside the largest five per input are set to 0. The weights
947 are initialized using Xavier/Glorot. The training is conducted at mixed precision (fp16), with the last
948 ten epochs performed at full precision.

949 **Optimizer and Schedule.** The model is trained for 50 epochs using base AdamW ($\beta_1 = 0.9$,
950 $\beta_2 = 0.999$) optimizer is employed with weight decay set to 10^{-5} . Linear warm-up is applied on the
951 first 5% of steps, followed by cosine decay from $\eta_{\max} = 5 \times 10^{-4}$ to $\eta_{\text{final}} = 10^{-6}$. MSE loss is used
952 alongside an auxiliary term penalizing activations that never enter the top- k set, where $\lambda = 10^{-5}$.

953 C Computational Resources

954 This section summarizes the GPU resources used for training and experiments in support of this
955 paper. In total, we used approximately 202 GPU-hours on NVIDIA H100s and H200s.

956 **RA-SAE.** Trained for approximately 24 GPU-hours on three NVIDIA H100s.

957 **Synthesized Images.** Generating the full $D_{g\theta}$ (see Appendix A) took roughly 5 hours per generator
958 when distributed across four NVIDIA H200 GPUs. With five generators, this totaled approximately
959 100 GPU-hours on a NVIDIA H200.

960 **● Distribution-Level Analysis.** Extracting energy differences at the distribution level took about 3
961 hours per generator on a single NVIDIA H200 GPU (total ~ 15 GPU-hours).

962 **● Datapoint-Level Analysis.** Computing datapoint-level energy differences, ranking concepts per
963 datapoint, and ranking datapoints per concept also took approximately 3 hours per generator on one
964 H200 GPU (total ~ 15 GPU-hours).

D Autointerpretability

To automate the interpretation of concepts identified by the RA-SAE, we qualitatively evaluated how well Vision-Language Models (VLMs) describe them. We queried ChatGPT 4o via the OpenAI API using various prompts and formats of an image x'' in which the given concept was salient, recording the VLM’s description. This section reports our findings. For evaluation we let the VLM provide its full reasoning; a scalable deployment need only modify the prompt—asking for the description alone—to filter out the reasoning.

First, we provided the VLM with two images: a raw version of x'' and x'' with a heatmap superimposed to localize the intensity of the concept:

Input



You are given two images: (1) raw image and (2) the same image with a heatmap of areas that are salient for a concept present in the image, as identified by a DiNOv2-based SAE. Describe what is this concept.

Output

The concept identified by the DiNOv2-based SAE in these images appears to correspond to **"ring"** or **"metal ring"** as the salient object.

Reasoning:

- In the original image, a sleek, brushed-metal ring is centered on a wooden surface.
- In the heatmap overlay, the regions that are most activated (least transparent with bright color blocks) are tightly focused on the ring itself, particularly the outer surface and inner reflections.
- The wooden background is mostly ignored except for minor scattered activations, confirming that the concept is not about the background texture.
- The consistent and strong focus on the metallic, circular object suggests that the model recognizes this as the defining element—likely trained to detect the shape, material, and reflectivity typical of rings.

As the example shows, the VLM misinterpreted the heatmap visualization: instead of describing the wood texture in the background—the actual salient region—it described the foreground ring, which received no attention from the RA-SAE.

To determine whether this was merely a prompting issue, we repeated the same scenario with an expanded prompt that explained the heatmap in greater detail:

Input



You are given two images: (1) raw image and (2) the same image with a superimposed heatmap of areas that contain a concept identified by a DiNOv2-based SAE. Red areas are the most salient (contain the concept); blue areas are the least salient (do not contain the concept). Describe what is this concept.

Output

The concept identified by the DiNOv2-based SAE in these images appears to be **"ring"** or more specifically **"metal band/jewelry ring."**

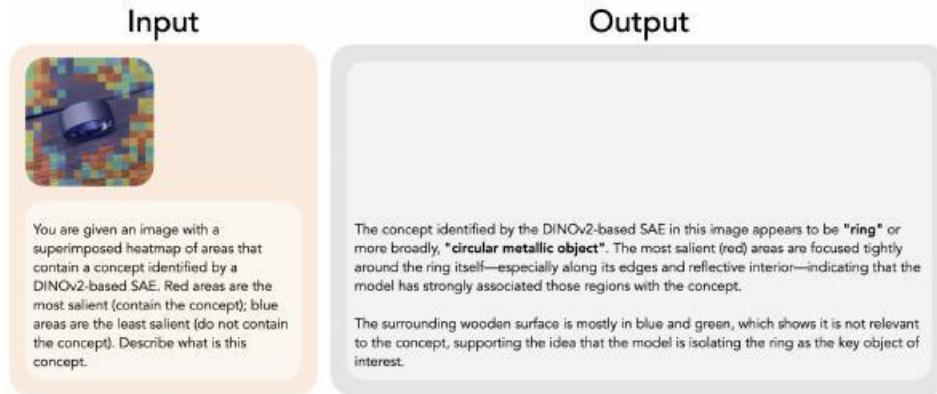
Here's the reasoning:

- In the heatmap image, the red and yellow (most salient) regions align very closely with the circular metal ring at the center of the image.
- The ring is the only distinct object in the photo and is clearly segmented from the background (wooden surface), which is mostly marked in blue and green (least salient).
- The attention does not focus on the texture of the background or lighting but is highly localized to the curved, metallic object.

Thus, the concept is almost certainly the metallic ring—its shape, reflectivity, and material seem to trigger the sparse autoencoder's activation, making it a learned concept in the SAE dictionary.

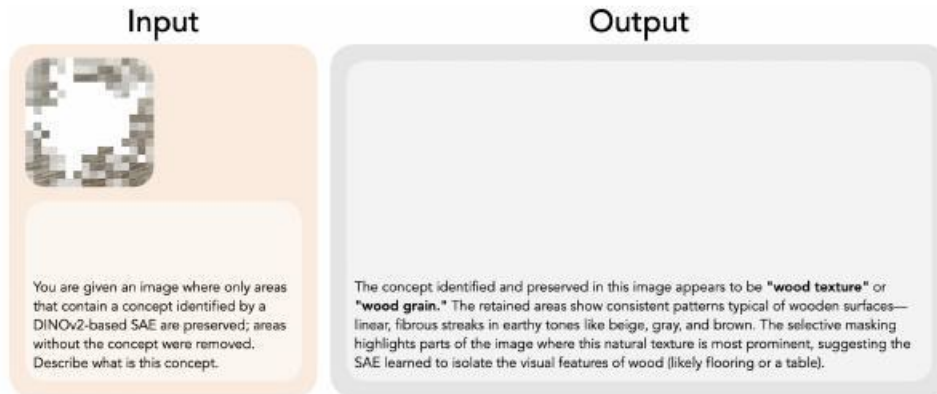
This modification did not resolve the issue: the VLM still identified the ring as the prominent concept. Several further prompting techniques, including ones that described the heatmap in even greater depth (e.g., by including the code used to generate it), produced the same result.

982 We next tested providing the VLM only with the heatmap-superimposed x'' and a similar prompt
 983 explaining how to interpret the heatmap:



984 Once again, the VLM failed to identify the wooden background and instead focused on the ring. As
 985 before, various prompting strategies with this image format left the results essentially unchanged.

986 These observations establish that the heatmap alone cannot reliably guide the VLM's attention to the
 987 salient regions; the presence of less relevant regions distracts the model. We therefore replaced the
 988 heatmap-superimposed x'' with an alpha-masked version of x'' :



989 As the example shows, the VLM now correctly identified wood texture as the dominant concept,
 990 and we observed consistent success across a range of concepts and images.

991 The formatting of x'' and the prompt shown above yielded the most reliable results in our qualitative
 992 evaluation. This evaluation is bounded by its qualitative nature (there are no ground-truth annotations)
 993 and its focus on a single VLM; we expect future work on the autointerpretability of SAE concepts to
 994 build directly on this analysis.

995 **E Additional Results: Model-Specific Blindspots**

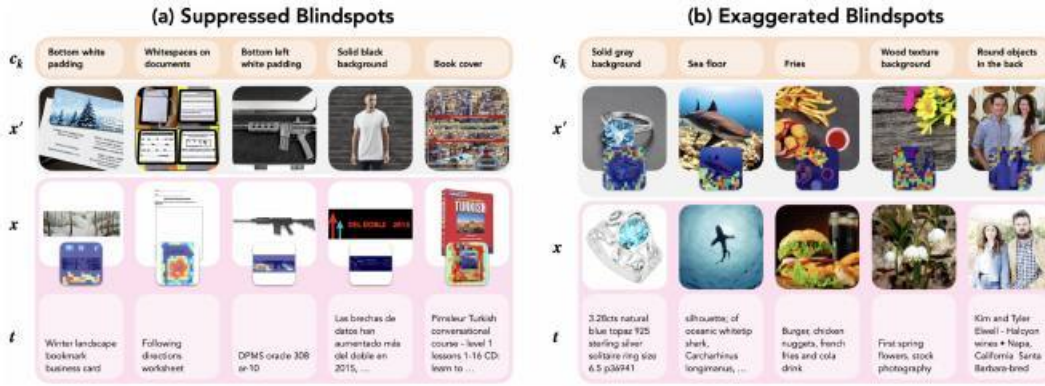


Figure 14: Conceptual blindspots in **Stable Diffusion 1.5**. For each concept, we show the prototypical natural (for suppressed blindspots) or synthesized (for exaggerated blindspots) image, selected by highest absolute activation, with the concept’s spatial heatmap superimposed.

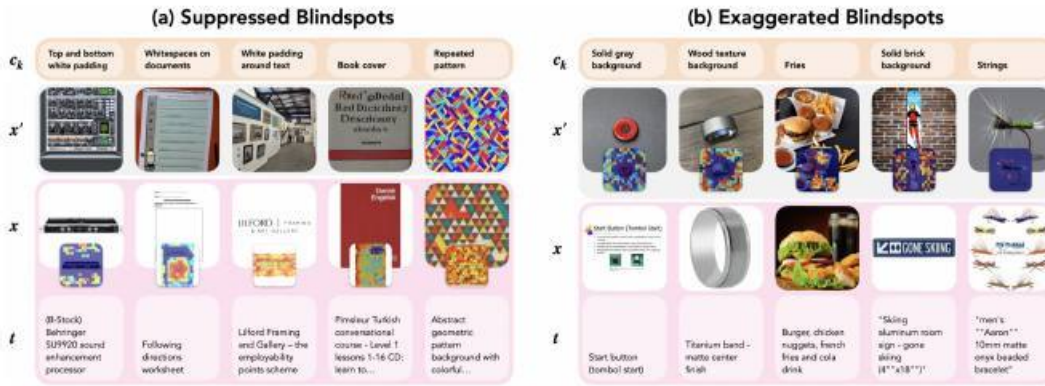


Figure 15: Conceptual blindspots in **Stable Diffusion 2.1**. For each concept, we show the prototypical natural (for suppressed blindspots) or synthesized (for exaggerated blindspots) image, selected by highest absolute activation, with the concept’s spatial heatmap superimposed.

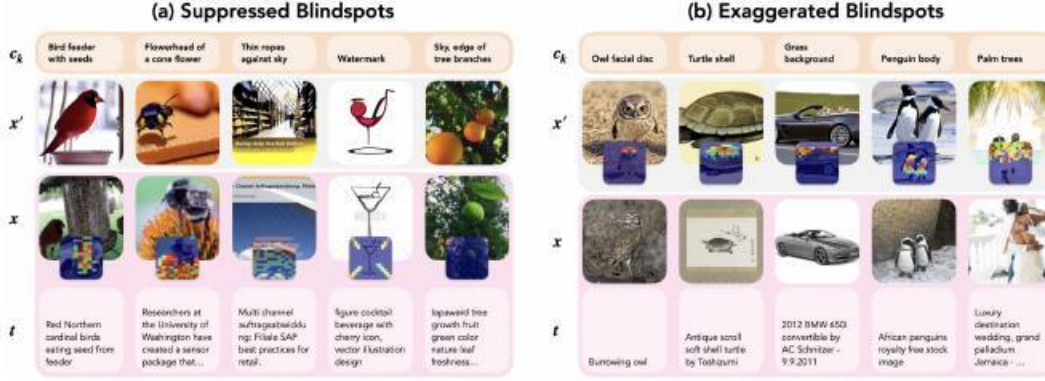


Figure 16: Conceptual blindspots in **Kandinsky**. For each concept, we show the prototypical natural (for suppressed blindspots) or synthesized (for exaggerated blindspots) image, selected by highest absolute activation, with the concept's spatial heatmap superimposed.



Figure 17: Conceptual blindspots in **PixArt**. For each concept, we show the prototypical natural (for suppressed blindspots) or synthesized (for exaggerated blindspots) image, selected by highest absolute activation, with the concept's spatial heatmap superimposed.

F Additional Results: Higher-order Blindspots with Compositional Discrepancy

Thus far, our analysis has centered on individual concept activations. Yet visual scenes are rarely composed of isolated concepts; they are structured through rich co-occurrence patterns that encode compositional semantics. We now determine whether generative models capture this higher-order structure by analyzing the co-activation matrix $Z^T Z$, which encodes the pairwise correlations in concept usage.

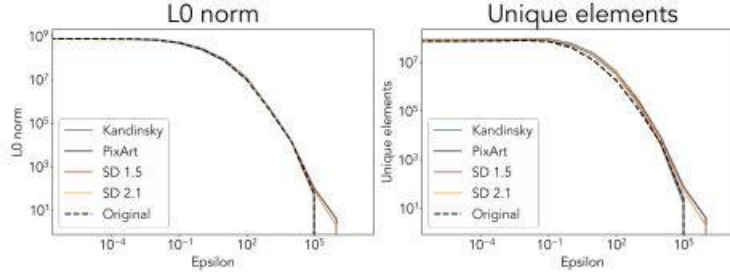


Figure 18: • **Sparsity and Structural Divergence.** Left: L_0 norm of the co-occurrence matrix $Z Z^T$ as a function of the threshold ϵ , showing how many entries remain active in each model. Right: number of unique entries in the synthesized distribution relative to the natural distribution. Every evaluated model preserves the global sparsity structure yet diverges in activation content.

Strikingly, at the level of binary structure diffusion models approximate the global sparsity of the natural co-occurrence matrix with high fidelity. As Fig. 18 (left) shows, the ℓ_0 norm of $Z^T Z$ —thresholded at varying ϵ values—tracks closely between the natural and synthesized distributions across all models. The gross connectivity of the conceptual graph—which concepts co-activate at all—is thus well preserved. Formally, $Z^T Z$ is the adjacency matrix of a weighted, undirected graph over concepts, with edge weights reflecting co-activation strength across the dataset.

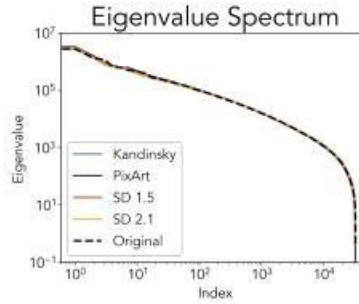


Figure 19: • **Spectral Structure of Co-occurrence.** Log-log plot of the eigenvalue spectra of the co-occurrence matrices $Z Z^T$ across models and the natural distribution. Every evaluated model matches the heavy-tailed decay of the natural distribution.

The specific content of these co-activations, however, diverges sharply: as Fig. 18 (right) illustrates, a substantial portion of the entries in the model-generated $Z^T Z$ are absent from the natural baseline. The capacity for compositionality is therefore retained even as the identity of active pairings shifts—a shift that reflects model-specific inductive biases or training artifacts. To probe the internal structure of these co-occurrence patterns, we turn to spectral analysis. Fig. 19 shows the eigenvalue spectra of the co-occurrence matrices for each model and the natural distribution. Every spectrum exhibits a heavy-tailed, power-law decay, establishing that generative models preserve the overall rank structure and variance allocation across conceptual dimensions.

We further examine the alignment of dominant conceptual directions via cosine similarity heatmaps between the top 100 eigenvectors of the synthesized and natural co-occurrence matrices (Fig. 20). All models exhibit partial diagonal alignment—overlap in the principal concept axes—but the off-

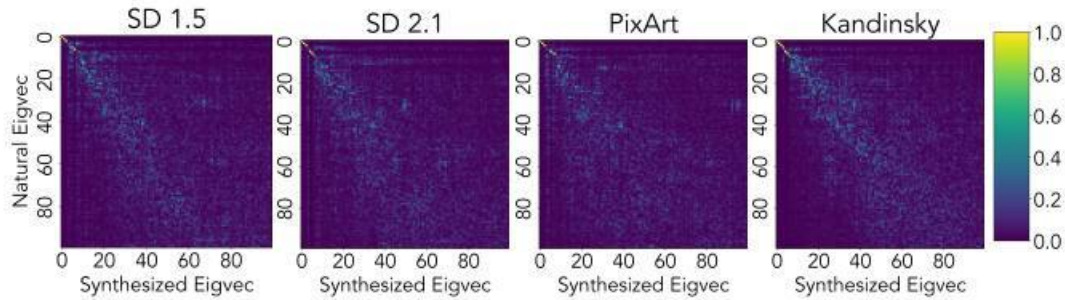


Figure 20: ● **Concept Basis Similarity.** Cosine similarity heatmaps between the top 100 eigenvectors of the natural and synthesized co-occurrence matrices ZZ^T . The diagonal structure marks alignment of the dominant conceptual directions, whose strength varies clearly across the four models.

1021 diagonal entries expose clear rotations and mismatches in higher modes, marking definite deviations
1022 in compositional geometry.

1023 Together, these findings establish that diffusion models approximate the global shape of concept
1024 co-activation remarkably well, yet deviate in subtle, structured ways under the spectral lens. These
1025 higher-order discrepancies plausibly underpin the observed failures in generating coherent, multi-
1026 object scenes and relational concepts.

1027 G Additional Results: Blindspot Stress Testing

1028 To stress-test the blindspots identified by our method, we gathered a range of prompts describing
 1029 these blindspots and used them to generate many images. We then directly contrasted the outputs of
 1030 models in which the concept was identified as a blindspot against those in which it was not.

1031 Specifically, ChatGPT-4o was prompted as follows: *I want to generate an image of the following*
 1032 *concept: "<blindspot>". Suggest 50 prompts highlighting this concept to be used as input for a*
 1033 *text-to-image model. Return these as a list of strings in Python.* Five images were generated per
 1034 prompt and analyzed with our custom RA-SAE model (see Appendix B), which ranked them by
 1035 the intensity with which the desired concept appeared. We then manually reviewed all images to
 1036 determine whether the blindspot was successfully depicted.

1037 As the following examples of suppressed concepts show, while some aspects of the target concept
 1038 occasionally surfaced (e.g., a holder or string for the bird feeder blindspot and a round hole for
 1039 the glossy DVD disc blindspot), the models consistently failed to generate the full concept. This
 1040 directly corroborates our method’s assessment and confirms the validity of the stress test.

1041 G.1 Bird Feeder Blindspot in Kandinsky



Figure 21: Images generated from various prompts involving the bird feeder concept during stress testing. Our method identified this concept as a suppressed conceptual blindspot in Kandinsky, and the behavior matches exactly: the model cannot generate a corresponding image. By contrast, SD 1.5—in which this concept was not identified as a blindspot—generates it successfully.



Figure 22: Images generated from various prompts involving the glossy DVD disc concept during stress testing. Our method identified this concept as a suppressed conceptual blindspot in SD 1.5, and the behavior matches exactly: the model cannot generate a corresponding image. By contrast, Kandinsky—in which this concept was not identified as a blindspot—generates it successfully.

1043 **H Additional Examples: Synthesized Images**



Figure 23: Additional LAION-5B image-caption pairs, paired with images generated from the same prompt by SD 1.5, SD 2.1, PixArt, and Kandinsky.

1044 I Concentration bounds for δ

1045 In our experiments, we estimate $\delta(k)$ from $n = 10,000$ paired samples per concept. This budget
 1046 is modest, which raises the natural question of whether it suffices for reliable estimates. We settle
 1047 the question by deriving a concentration bound on the empirical estimator $\hat{\delta}_n(k)$ via McDiarmid's
 1048 inequality [142]. The bound is tight and proves that even relatively few samples yield fast, accurate
 1049 estimates of concept bias.

1050 **Theorem 4** (Concentration of $\hat{\delta}_n(k)$). *We assume that the concept score $\xi_k(\mathbf{x})$ takes values in $[a, b]$
 1051 almost surely for all images $\mathbf{x}$ drawn from either $\mathcal{D}_X$ or $\mathcal{D}'_X$. Let n paired samples $(\mathbf{x}_i, \mathbf{x}'_i)_{i=1}^n$ be
 1052 drawn independently with $\mathbf{x}_i \sim \mathcal{D}_X$ and $\mathbf{x}'_i \sim \mathcal{D}'_X$, and define the empirical estimator*

$$\hat{\delta}_n(k) := \sigma \left(\frac{1}{n} \sum_{i=1}^n \xi_k(\mathbf{x}_i) - \frac{1}{n} \sum_{i=1}^n \xi_k(\mathbf{x}'_i) \right).$$

1053 Let $M := b - a$ and $L := M/4$. Then for every $\varepsilon > 0$, the deviation satisfies

$$\mathbb{P} \left(\left| \hat{\delta}_n(k) - \delta(k) \right| > \varepsilon \right) \leq 2 \exp \left(-\frac{2n\varepsilon^2}{L^2} \right).$$

1054 *Proof.* The function $x \mapsto \sigma(x)$ is $1/4$ -Lipschitz, since $|\sigma'(x)| \leq 1/4$ for all x .

1055 Viewing $\hat{\delta}_n(k)$ as a function of the $2n$ independent variables $(\mathbf{x}_1, \dots, \mathbf{x}_n, \mathbf{x}'_1, \dots, \mathbf{x}'_n)$, changing a
 1056 single argument alters the inner difference of means by at most M/n , and the outer sigmoid scales this
 1057 by at most $1/4$. Hence, the bounded difference constant for each coordinate is $(M/n)(1/4) = L/n$.

1058 By McDiarmid's inequality [142],

$$\mathbb{P} \left(\left| \hat{\delta}_n(k) - \delta(k) \right| > \varepsilon \right) \leq 2 \exp \left(-\frac{2\varepsilon^2}{\sum_{j=1}^{2n} (L/n)^2} \right) = 2 \exp \left(-\frac{2n\varepsilon^2}{L^2} \right),$$

1059 which proves the claim. □

1060 In practice, most concept scores $\delta(k)$ are sparse, with the majority concentrated near zero and only a
 1061 few reaching values up to 10. The concentration bound shows that even for the largest observed biases,
 1062 a sample size of $n = 10,000$ keeps $\hat{\delta}_n(k)$ within a small ε of the true value with high probability.
 1063 This directly justifies our sampling strategy and establishes that accurate bias measurements are
 1064 attainable with limited data.

1065 J Monotonicity and Calibration-Free Interpretation of δ

1066 Our goal when analyzing blind spots is to rank concepts by the severity of their generative bias, and
 1067 we use the score $\delta(k)$ for this purpose. One might worry that this score introduces distortions relative
 1068 to more direct quantities such as the energy gap or the odds ratio. The following result removes that
 1069 concern: $\delta(k)$ is a strictly increasing reparameterization of both and therefore inherits their exact
 1070 ordering, so no calibration whatsoever is needed when using $\delta(k)$ to rank concepts.

1071 **Theorem 5** (Monotonicity and Calibration of $\delta_{g_\theta \leftrightarrow G}$). *For every concept index k define the energy*
 1072 *gap*

$$\Delta_k = \mathbb{E}_{\mathbf{x}' \sim \mathcal{D}'_k}[\xi_k(\mathbf{x}')] - \mathbb{E}_{\mathbf{x} \sim \mathcal{D}_k}[\xi_k(\mathbf{x})],$$

1073 *the associated odds ratio $\rho_k = \exp(\Delta_k)$, and the energy-difference score*

$$\delta(k) = \frac{1}{1 + \exp(-\Delta_k)} = \frac{\rho_k}{1 + \rho_k}.$$

1074 *Then $\delta(k)$ is a strictly increasing bijection of both Δ_k and ρ_k , so ranking concepts by any one of*
 1075 *$\delta(k)$, Δ_k , or ρ_k produces exactly the same ordering.*

1076 *Proof.* The logistic sigmoid satisfies $\sigma'(x) = \sigma(x)(1 - \sigma(x)) > 0, \forall x \in \mathbb{R}$; hence σ and therefore
 1077 $\delta(k) = \sigma(\Delta_k)$ grow strictly with Δ_k . Because the exponential map is also strictly increasing and
 1078 bijective $\mathbb{R} \rightarrow (0, \infty)$, setting $\rho_k = \exp(\Delta_k)$ preserves order and gives $\Delta_k = \log \rho_k$. Substituting
 1079 this identity into σ yields $\delta(k) = \sigma(\log \rho_k) = \rho_k / (1 + \rho_k)$, which is the composition of two
 1080 strictly increasing bijections and is therefore itself strictly increasing and bijective in ρ_k . Since strict
 1081 monotonic functions never reverse inequalities, the three quantities share the same total order over
 1082 concepts.

1083

□

1084 Thus, ranking concepts by $\delta(\cdot)$ is strictly equivalent to ranking them by energy gap or by conceptual
 1085 generation odds ρ_k : no calibration is necessary, and all three quantities preserve exactly the same
 1086 total ordering over concepts.

1087 K Stability of FID Under SAE Embeddings

1088 In this section we establish a quantitative relationship between the Fréchet Inception Distance
 1089 (FID) computed in the original activation space (of dimension d) and the FID after applying a
 1090 (potentially overcomplete) SAE dictionary $D \in \mathbb{R}^{k \times d}$ with $k \gg d$. Throughout we assume that D
 1091 has orthonormal columns but is not necessarily square, i.e.

$$D^\top D = I_d, \quad \text{while} \quad DD^\top \neq I_k.$$

1092 We start from a simple fact: if D is not overcomplete, orthogonal, and $k = d$, then A and Z are
 1093 related by an isometry, so FID is preserved exactly. This case is unrealistic, however, so we turn to
 1094 the overcomplete case and show that FID is bounded by the extreme singular values of D . We work
 1095 with the Wasserstein-2 metric $\mathcal{W}_2$, using that FID is precisely $\mathcal{W}_2^2$ specialised to Gaussians.

1096 For a probability measure μ on $\mathbb{R}^d$ we write $D_\# \mu$ for its push-forward under D , i.e. $D_\# \mu(z) =$
 1097 $\mu(D^{-1}z)$. Denote by $\sigma_{\min}$ and $\sigma_{\max}$ the minimal and maximal singular values of D , equivalently
 1098 the square-roots of the extremal eigenvalues of $DD^\top$:

$$\sigma_{\min}^2 I_k \preceq DD^\top \preceq \sigma_{\max}^2 I_k.$$

1099 Empirically one finds $\sigma_{\min}, \sigma_{\max} \approx 1$, but the proof relies on no such assumption. We begin with a
 1100 simple lemma for the case where D is not overcomplete.

1101 **Lemma 1** (Isometry under exact orthogonality). *Suppose $k = d$ and $D^\top D = DD^\top = I_d$. Then*
 1102 *D is an isometry: $\|Dv\|_2 = \|v\|_2$ for all $v \in \mathbb{R}^d$. Consequently, for any probability measures μ, ν*
 1103 *on $\mathbb{R}^d$ with finite second moment,*

$$\mathcal{W}_2(D_\# \mu, D_\# \nu) = \mathcal{W}_2(\mu, \nu).$$

1104 *Proof.* Orthogonality of D implies preservation of the Euclidean norm, and push-forward commutes
 1105 with the map inside the $\mathcal{W}_2$ infimum; the integrand is unchanged, so the infimum value is identical.
 1106 □

1107 This case, however, is quite unrealistic, since SAEs rely on overcompleteness to extract meaningful,
 1108 interpretable concepts. In the overcomplete case D is no longer orthonormal, but it can still be
 1109 column-orthonormal; we exploit this to bound the distortion by the extremal singular values of $D^\top D$.

1110 **Theorem 6** (FID under column orthogonal embeddings). *Let $D \in \mathbb{R}^{k \times d}$ satisfy $D^\top D = I_d$ and*
 1111 *denote by $0 < \sigma_{\min} \leq \sigma_{\max}$ the extreme singular values of $DD^\top$. Given two data matrices*
 1112 *$A, A' \in \mathbb{R}^{n \times d}$ (rows are sample vectors), set $Z = AD^\top \in \mathbb{R}^{n \times k}$ and $Z' = A'D^\top \in \mathbb{R}^{n \times k}$. Then*

$$\sigma_{\min}^2 \text{FID}(A, A') \leq \text{FID}(Z, Z') \leq \sigma_{\max}^2 \text{FID}(A, A').$$

1113 *Proof.* Write μ for the empirical measure of A and ν for that of A' , i.e.

$$\mu = \frac{1}{n} \sum_{i=1}^n \delta_{A_{i,:}}, \quad \nu = \frac{1}{n} \sum_{i=1}^n \delta_{A'_{i,:}}.$$

1114 For any coupling $\pi \in \Pi(\mu, \nu)$ (i.e. a probability measure on $\mathbb{R}^d \times \mathbb{R}^d$ with marginals μ, ν) we have,
 1115 by the extremal singular value bound,

$$\sigma_{\min}^2 \|x - y\|_2^2 \leq \|D(x - y)\|_2^2 \leq \sigma_{\max}^2 \|x - y\|_2^2, \quad \forall (x, y) \in \mathbb{R}^d \times \mathbb{R}^d.$$

1116 Integrating with respect to an arbitrary coupling $\pi \in \Pi(\mu, \nu)$ yields

$$\sigma_{\min}^2 \int \|x - y\|_2^2 d\pi \leq \int \|D(x - y)\|_2^2 d\pi \leq \sigma_{\max}^2 \int \|x - y\|_2^2 d\pi.$$

1117 The middle integral is exactly the transport cost of the pushed-forward coupling $(D \times D)_\# \pi$ between
 1118 $\mu_D := D_\# \mu$ and $\nu_D := D_\# \nu$. Because the inequalities hold for every π , they hold in particular
 1119 for the optimal couplings attaining $\mathcal{W}_2(\mu, \nu)$ and $\mathcal{W}_2(\mu_D, \nu_D)$, though these two optima need not

1120 coincide. Taking the infimum over π term-wise makes this explicit:

$$\sigma_{\min}^2 \inf_{\pi \in \Pi(\mu, \nu)} \int \|x - y\|_2^2 d\pi \leq \inf_{\pi \in \Pi(\mu, \nu)} \int \|D(x - y)\|_2^2 d\pi \leq \sigma_{\max}^2 \inf_{\pi \in \Pi(\mu, \nu)} \int \|x - y\|_2^2 d\pi.$$

1121 Hence

$$\sigma_{\min}^2 \mathcal{W}_2^2(\mu, \nu) \leq \mathcal{W}_2^2(\mu_D, \nu_D) \leq \sigma_{\max}^2 \mathcal{W}_2^2(\mu, \nu).$$

1122 Recognising $\text{FID}(\cdot, \cdot) = \mathcal{W}_2^2(\cdot, \cdot)$ for the Gaussian surrogate and plugging in (A, A') (resp. (Z, Z'))
 1123 finishes the proof. $\square$

1124 In short, Theorem 6 establishes that applying a column-orthogonal overcomplete SAE dictionary
 1125 cannot distort the Fréchet Inception Distance by more than the square of its extremal singular values.
 1126 When D is nearly orthogonal—empirically $\sigma_{\min}, \sigma_{\max} \approx 1$ —the result guarantees that FID measured
 1127 in the SAE feature space is essentially identical to the canonical FID. $\square$

1128